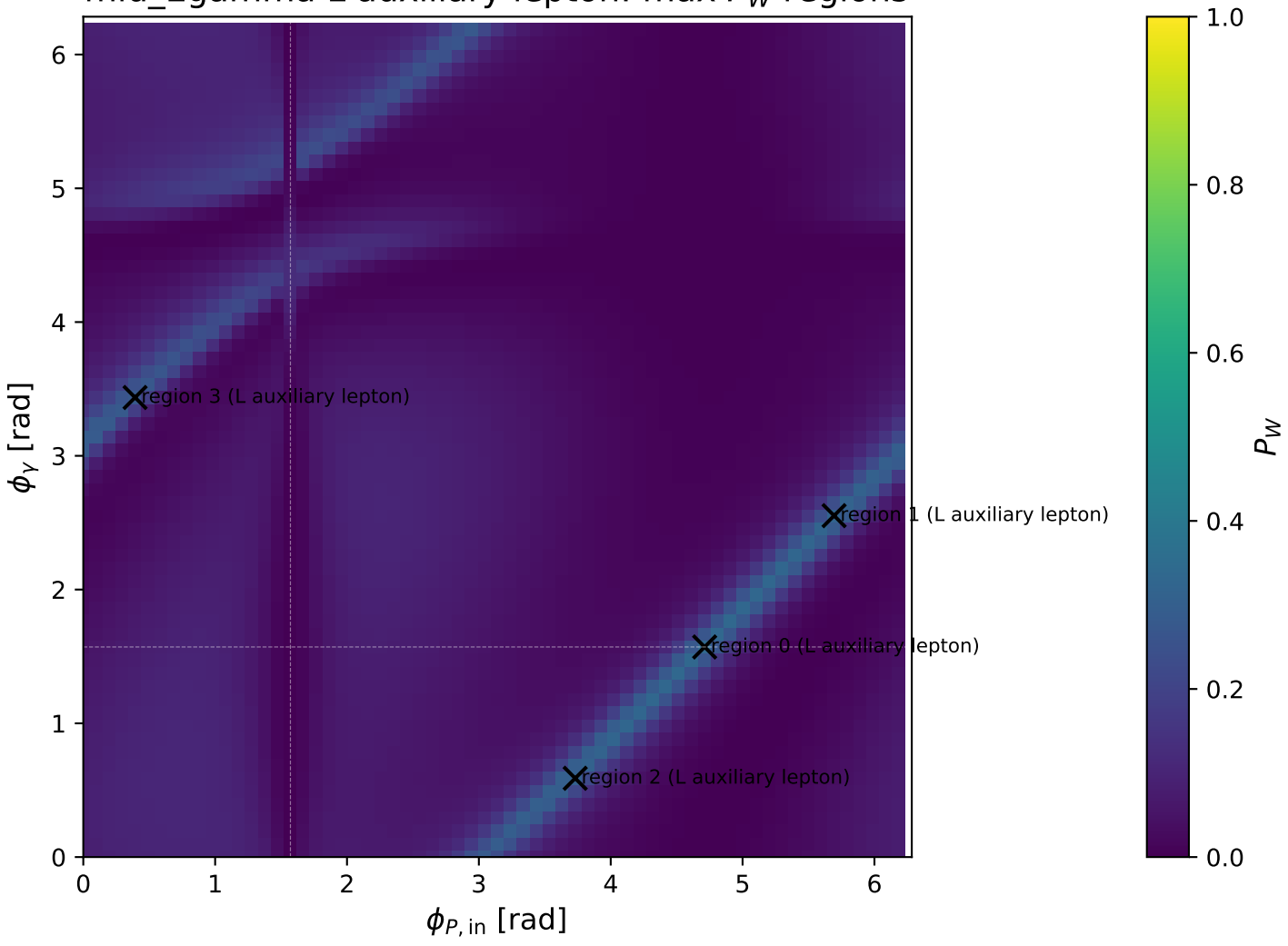
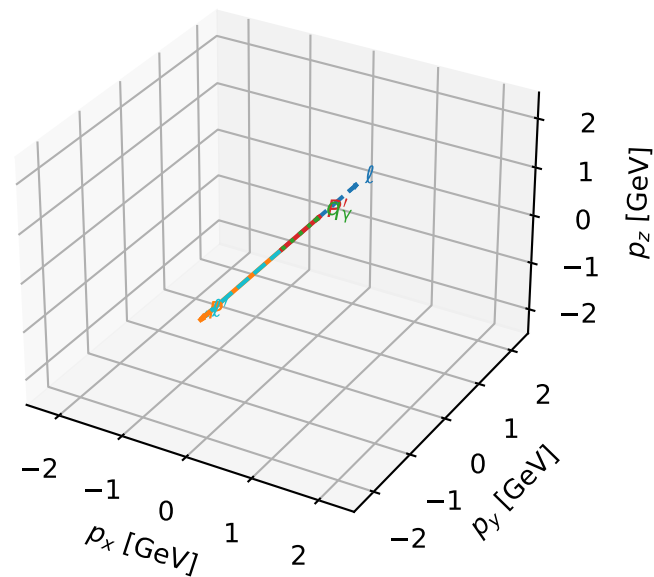


mid_Egamma L auxiliary lepton: max P_W regions



L auxiliary lepton characteristic max P_W configuration

3D momenta

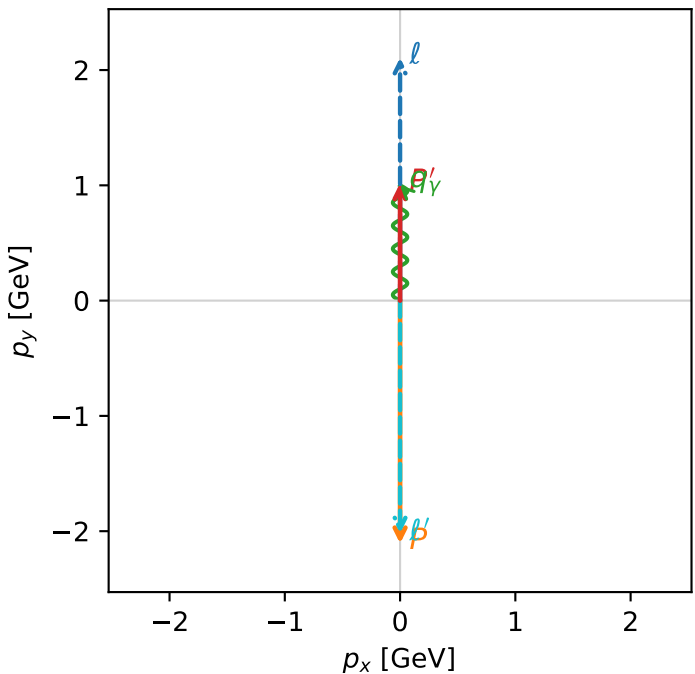


w_purity_mid egamma_l_electron_region_0 P_W=0.333333
region: mid_Egamma_L_electron_region_0
outgoing-state purity: 0.994471
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_l, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.01927$
 $\theta_{in}=1.57079$, $\phi_l=1.5708$, $\phi_P=4.71239$
 $E_\gamma=1$, $\phi_\gamma=1.5708$
 $Q^2=17.0194$, $x_B=0.958772$, $t=-8.92478$, $W^2=1.61168$, $y=0.904017$
 $\theta(l', \gamma)=180$ deg

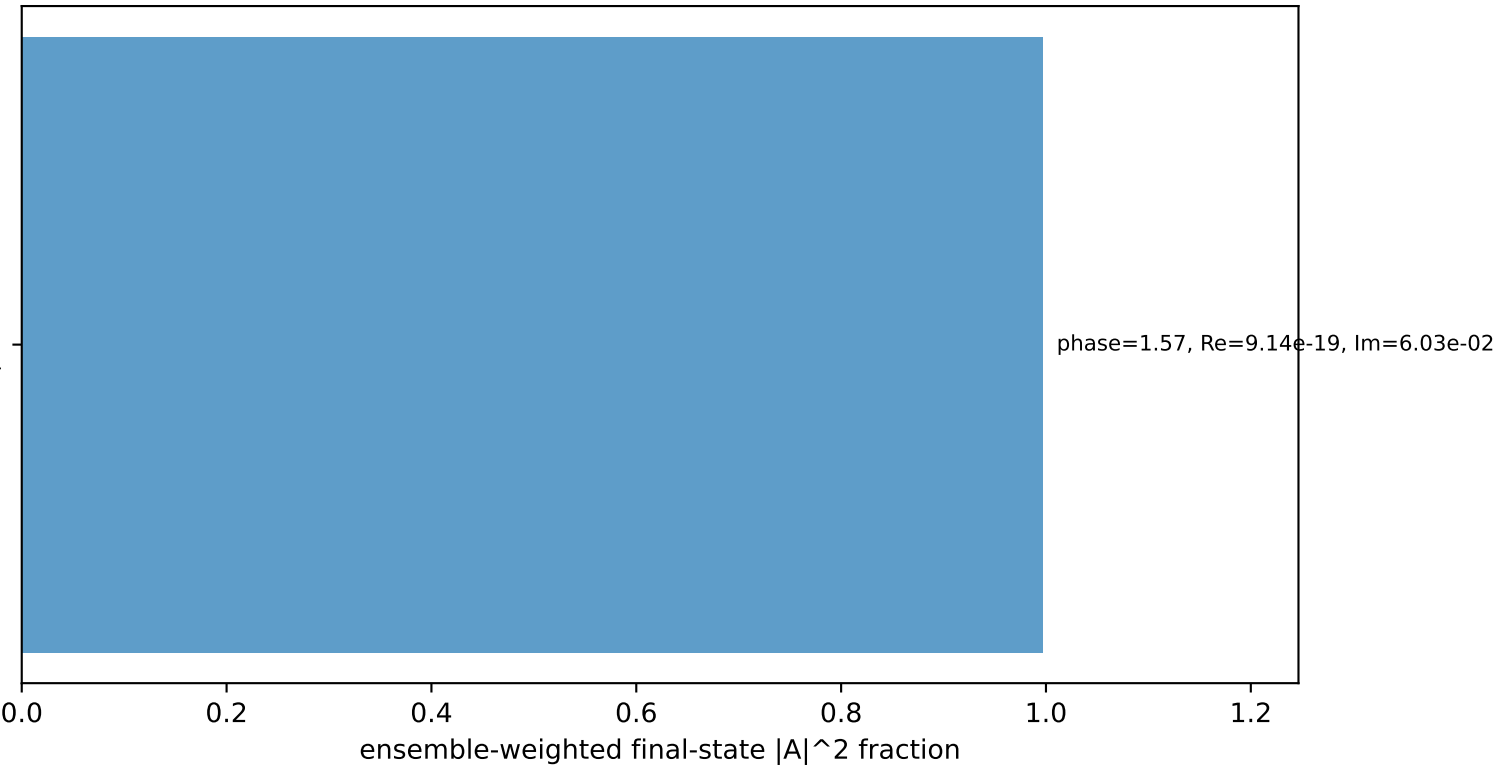
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= 0.0000$
 l' $E= 2.2533$ $p_x= -0.0000$ $p_y= -2.0193$ $p_z= 0.0000$
 P' $E= 1.3852$ $p_x= 0.0000$ $p_y= 1.0193$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.0000$ $p_y= 1.0000$ $p_z= 0.0000$

Transverse plane

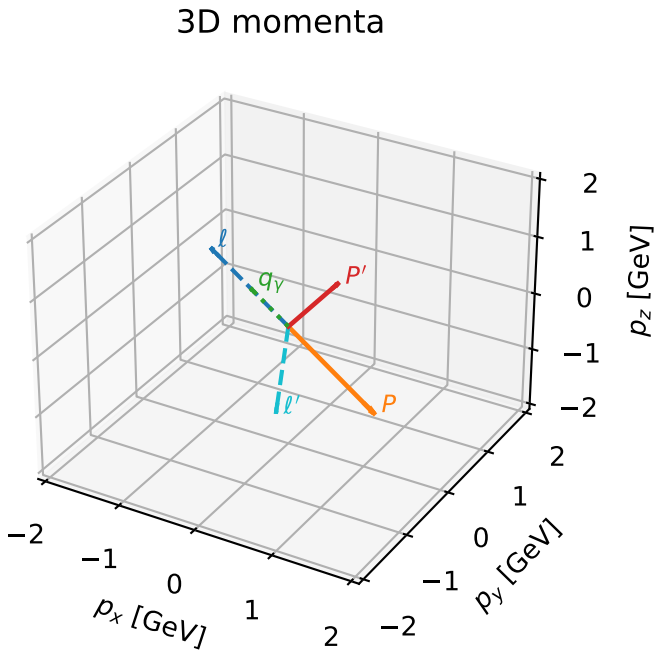


$h_e=-1, h_p=-1, h_\gamma=+1$
auxiliary lepton L, proton h=-1

Leading initial-ensemble/final-state components (N=1, retained=99.7%)



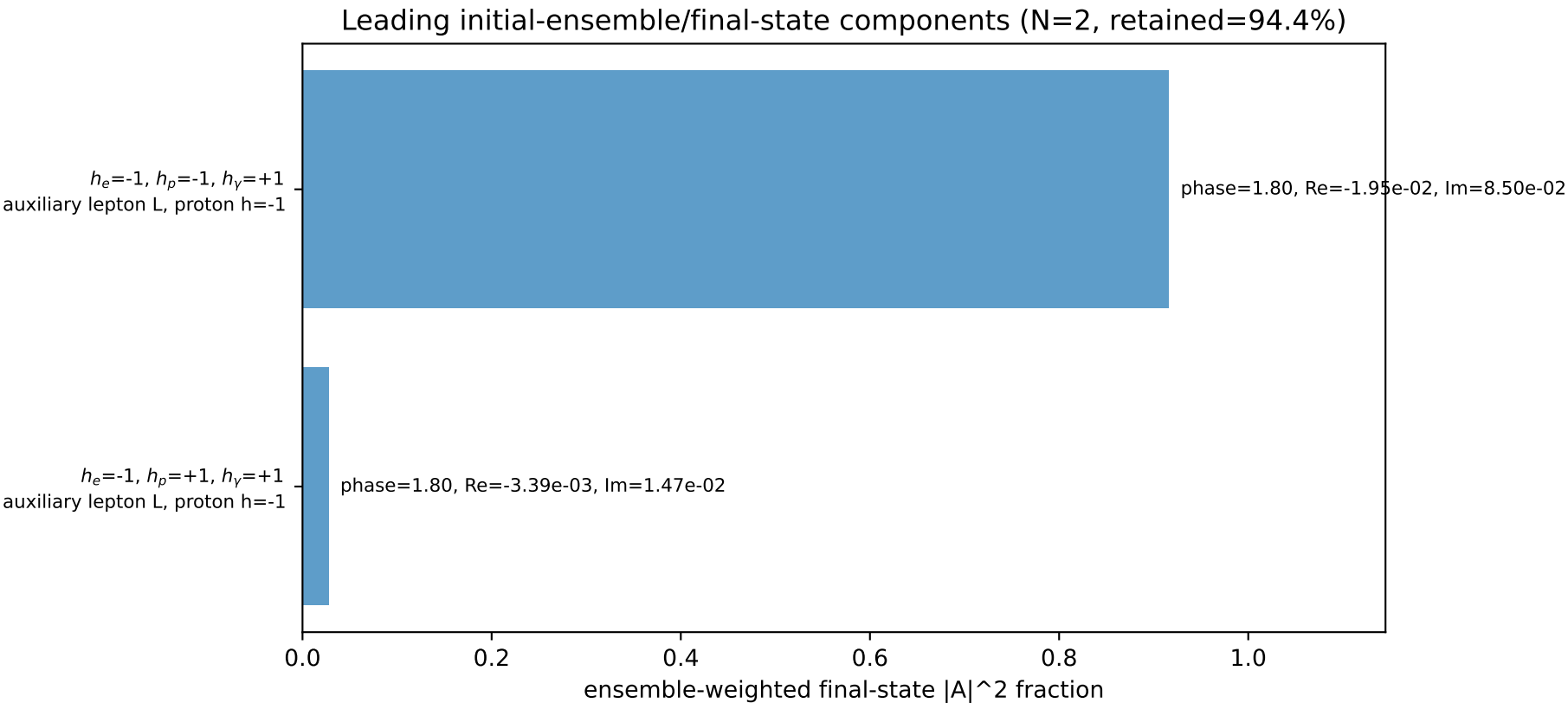
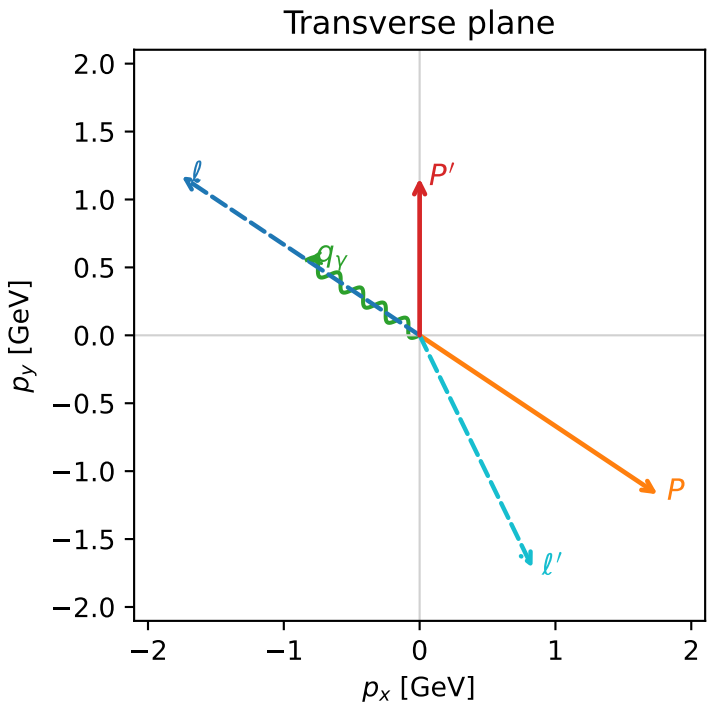
L auxiliary lepton characteristic max P_W configuration



w_purity_mid_egamma_l_electron_region_1 P_W=0.317785
region: mid_Egamma_L_electron_region_1
outgoing-state purity: 0.915787
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\ell, h_p)$

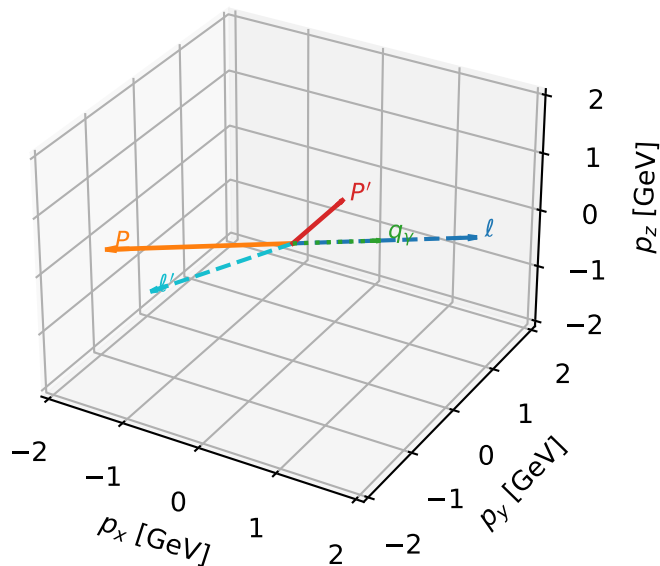
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.15614$
 $\theta_{in}=1.57079$, $\phi_\ell=2.55254$, $\phi_P=5.69414$
 $E_\gamma=1$, $\phi_\gamma=2.55254$
 $Q^2=14.9478$, $x_B=0.898264$, $t=-7.81421$, $W^2=2.5728$, $y=0.847459$
 $\theta(\ell', \gamma)=149.658$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -1.7519$ $p_y= 1.1706$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.7519$ $p_y= -1.1706$ $p_z= 0.0000$
 ℓ' $E= 2.1497$ $p_x= 0.8315$ $p_y= -1.7117$ $p_z= 0.0000$
 P' $E= 1.4888$ $p_x= 0.0000$ $p_y= 1.1561$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.8315$ $p_y= 0.5556$ $p_z= 0.0000$



L auxiliary lepton characteristic max P_W configuration

3D momenta

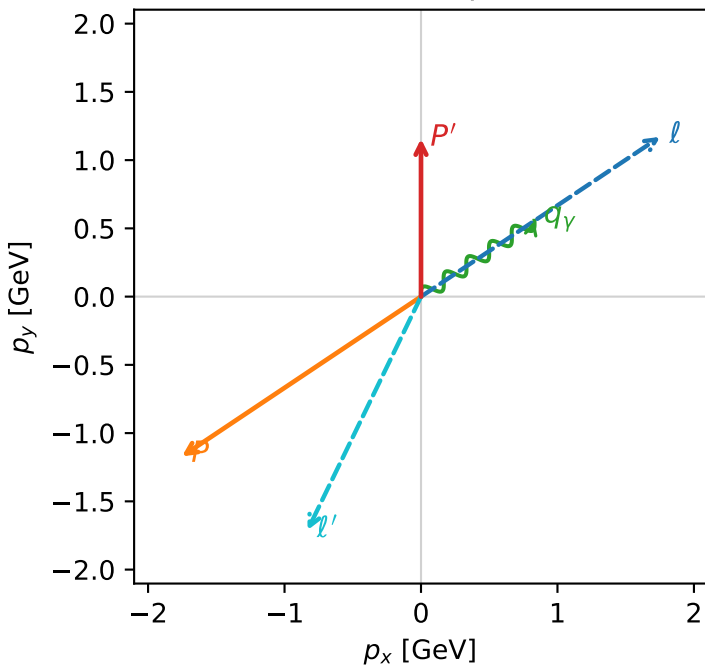


w_purity_mid_egamma_l_electron_region_2 P_W=0.30322
 region: mid_Egamma_L_electron_region_2
 outgoing-state purity: 0.915787
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_l, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.15614$
 $\theta_{in}=1.57079$, $\phi_l=0.589049$, $\phi_p=3.73064$
 $E_\gamma=1$, $\phi_\gamma=0.589049$
 $Q^2=14.9478$, $x_B=0.898264$, $t=-7.81421$, $W^2=2.5728$, $y=0.847459$
 $\theta(l', \gamma)=149.658$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 1.7519$ $p_y= 1.1706$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -1.7519$ $p_y= -1.1706$ $p_z= 0.0000$
 l' $E= 2.1497$ $p_x= -0.8315$ $p_y= -1.7117$ $p_z= 0.0000$
 P' $E= 1.4888$ $p_x= 0.0000$ $p_y= 1.1561$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.8315$ $p_y= 0.5556$ $p_z= 0.0000$

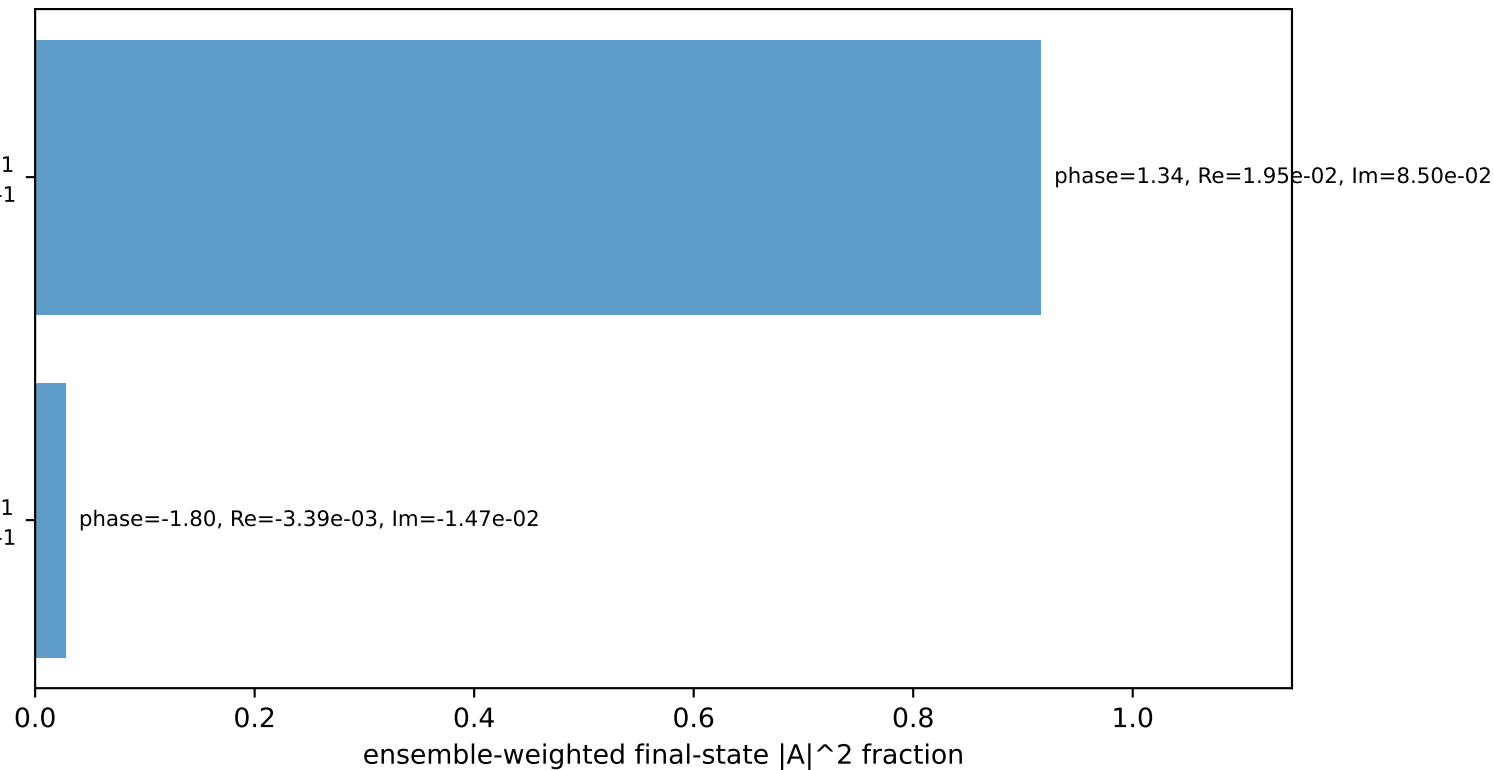
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
 auxiliary lepton L, proton h=-1

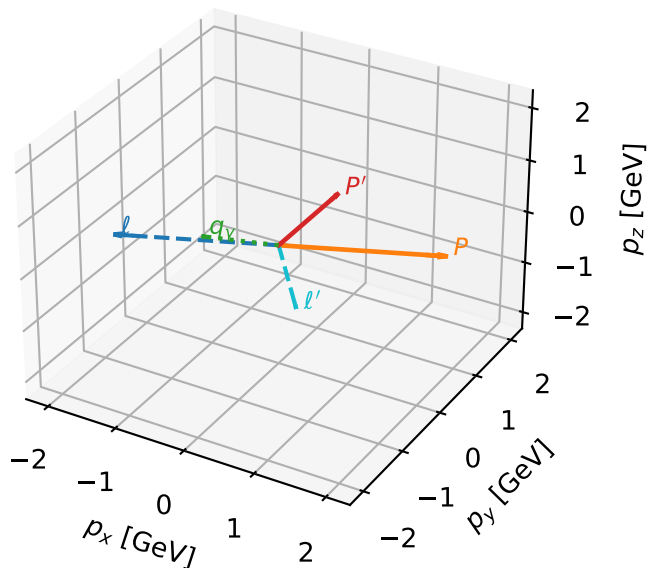
$h_e=-1, h_p=+1, h_\gamma=+1$
 auxiliary lepton L, proton h=-1

Leading initial-ensemble/final-state components (N=2, retained=94.4%)



L auxiliary lepton characteristic max P_W configuration

3D momenta

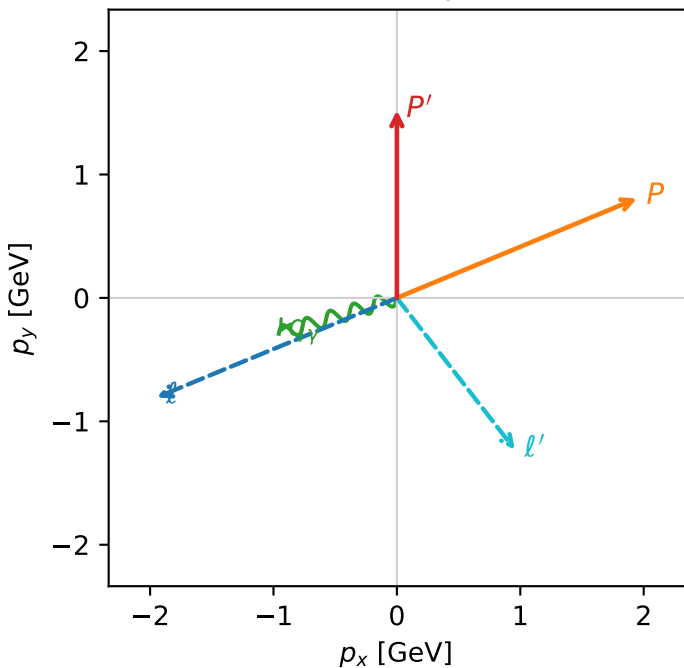


w_purity_mid_egamma_l_electron_region_3 P_W=0.27701
 region: mid_Egamma_L_electron_region_3
 outgoing-state purity: 0.763643
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\ell, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.52061$
 $\theta_{in}=1.57079$, $\phi_\ell=3.53429$, $\phi_P=0.392699$
 $E_\gamma=1$, $\phi_\gamma=3.43612$
 $Q^2=8.3794$, $x_B=0.65283$, $t=-4.0293$, $W^2=5.33595$, $y=0.653672$
 $\theta(\ell', \gamma)=111.001$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -1.9466$ $p_y= -0.8063$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.9466$ $p_y= 0.8063$ $p_z= 0.0000$
 ℓ' $E= 1.8519$ $p_x= 0.9569$ $p_y= -1.2303$ $p_z= 0.0000$
 P' $E= 1.7866$ $p_x= 0.0000$ $p_y= 1.5206$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.9569$ $p_y= -0.2903$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=5, retained=98.4%)

